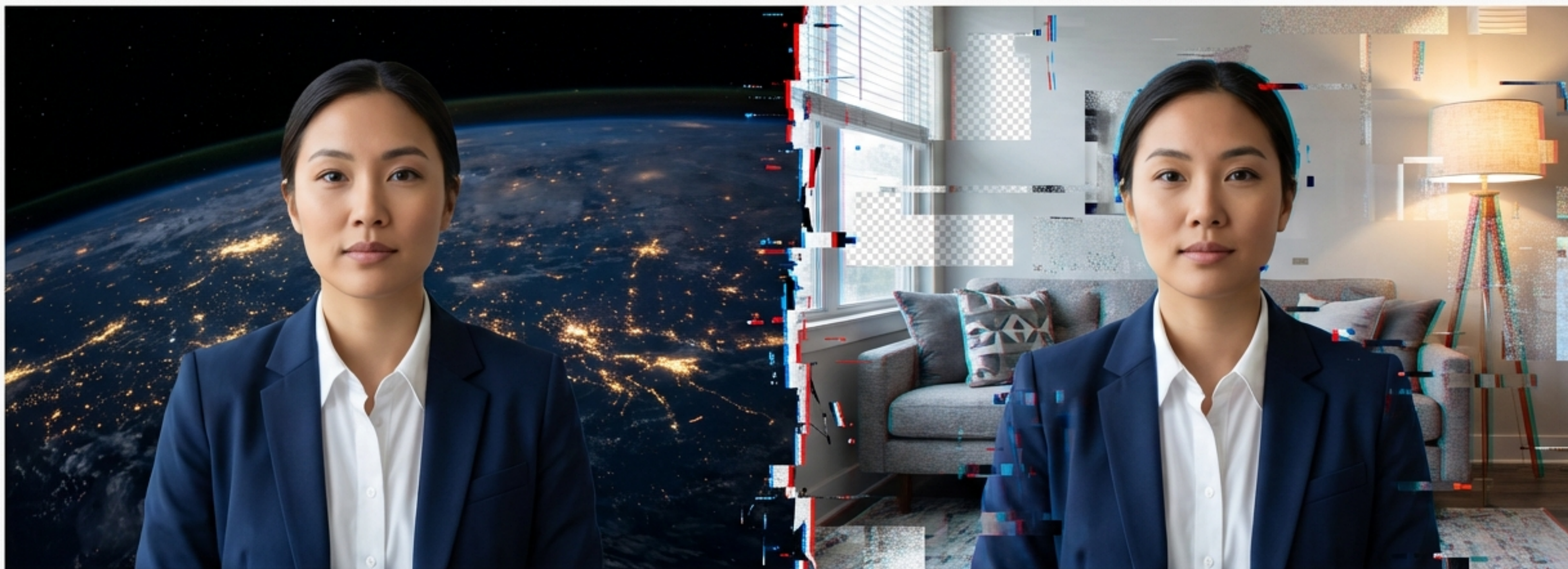


BEHIND THE VIRTUAL CURTAIN

An Investigative Report on the True Risks of Video Call Backgrounds



Based on research by Sabra et al., Nowroozi et al., Weissberg et al., and Kagan et al.

VIRTUAL BACKGROUNDS ARE USED AS AN EXPLICIT PRIVACY FEATURE

Users widely adopt virtual backgrounds not just for aesthetics, but with the specific intent of concealing their private environment.



of users agree that hiding objects motivates their use of virtual backgrounds.



agree they use them to cover a messy background.

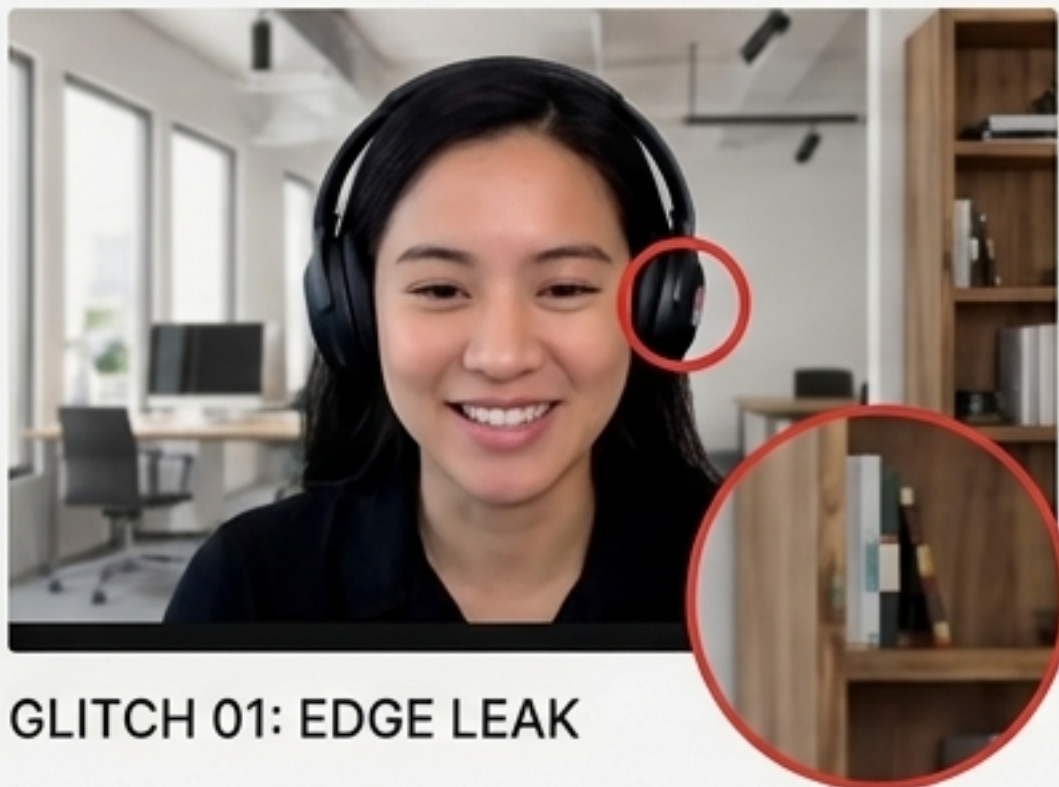
If this concealment fails due to an attack, **'84% of participants agree that this is an invasion to their privacy.'**

Source: Weissberg et al., "Seeing Through"

*This establishes a clear user expectation: **virtual backgrounds are a trusted privacy shield**. The stakes are high if that trust is misplaced.*

THE PRIVACY SHIELD IS A LEAKY VEIL

Seemingly minor glitches in virtual backgrounds are not random errors. They are consistent pixel leaks that can be aggregated over time to reconstruct significant portions of a user's private environment.



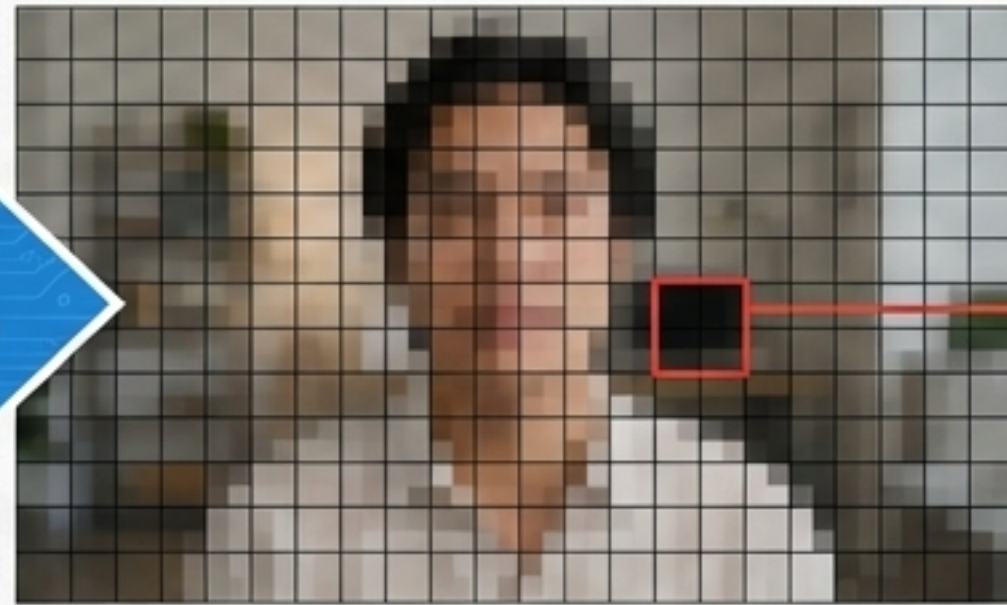
These leaks are the digital footprints that attackers can follow to see what's behind the curtain.

THE ROOT CAUSE: SEGMENTATION IS PERFORMED ON A LOW-RESOLUTION IMAGE

To ensure real-time performance on a wide range of hardware, video conferencing services don't segment the full-resolution video. They first shrink the frame, creating an inherent loss of detail.



Original Frame



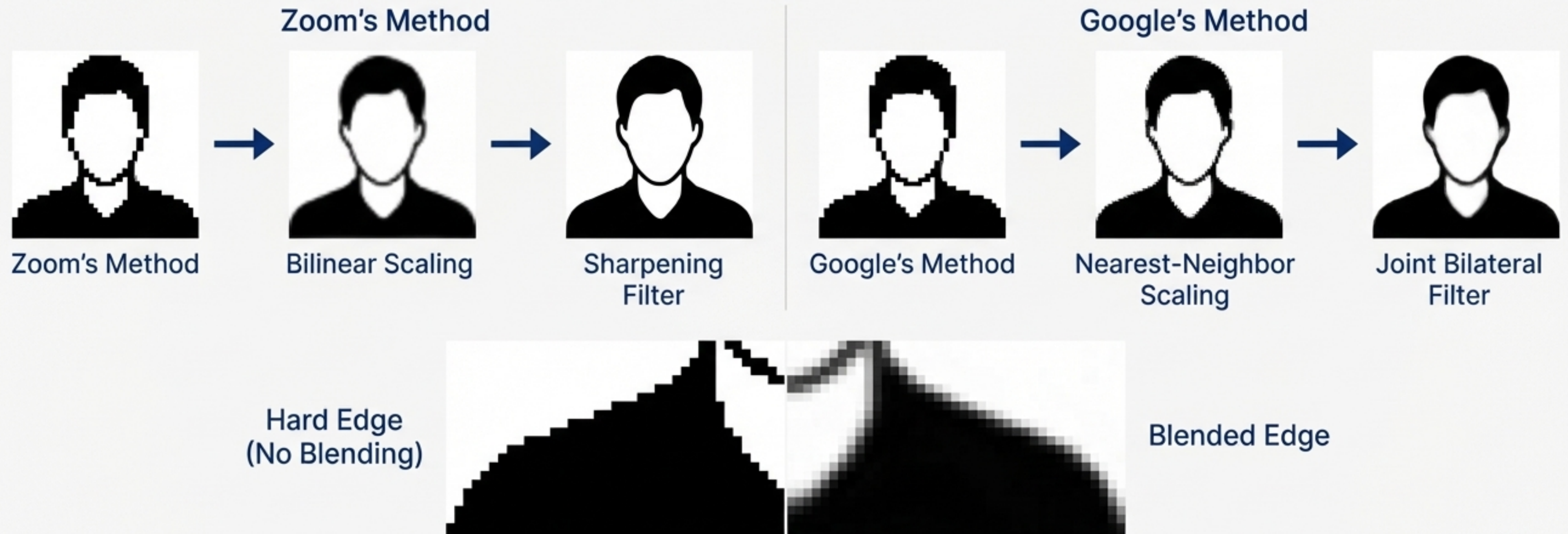
Segmentation Input (256x144)

The segmentation algorithm operates on a 256x144 pixel image. This means a single segmentation block corresponds to a **5x5 pixel area** in the original video. Any detail smaller than this is lost at the boundary, creating the initial leak.

Source: Weissberg et al., "Seeing Through"

SOPHISTICATED BLENDING MIXES REAL AND VIRTUAL PIXELS

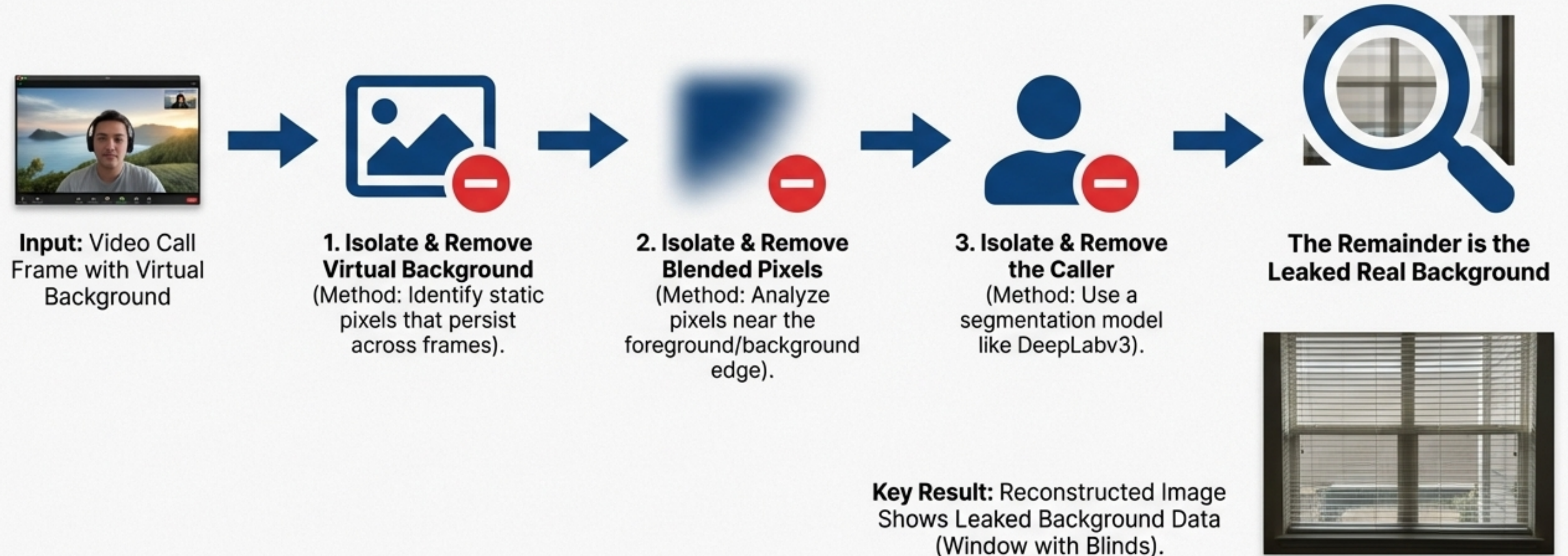
After the low-resolution mask is created, it's scaled back up. To hide the blocky artifacts, the software uses advanced blending filters. This process intentionally mixes pixels from the real background with the virtual one at the boundaries.



Blending creates a trail of evidence. The resulting pixels are neither purely from the real background nor the virtual one, but a complex mixture that an attacker can analyze to reverse-engineer the original scene.

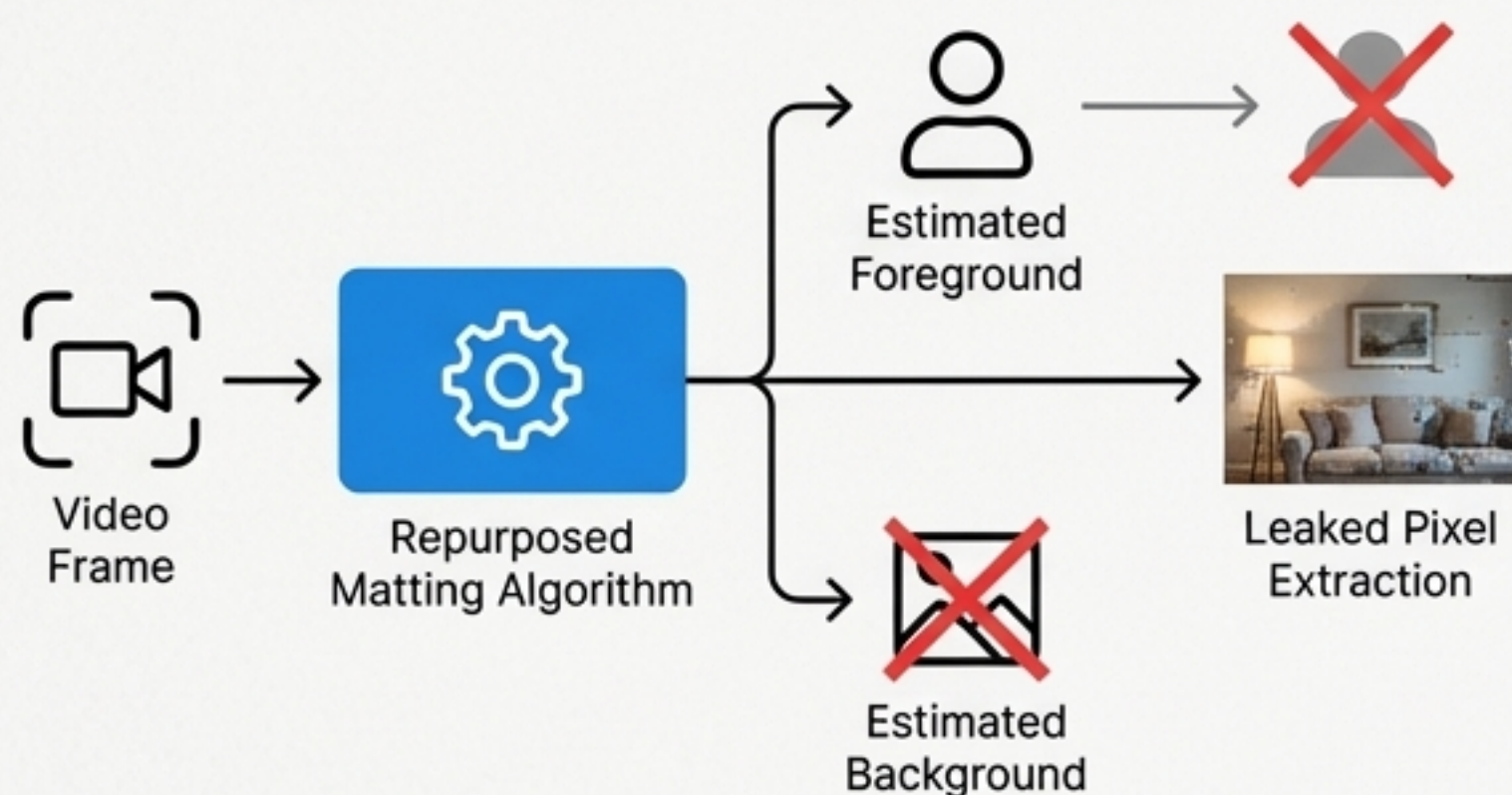
THE 'BACKGROUND BUSTER' FRAMEWORK: A PROCESS OF ELIMINATION

The first generation of attacks operates on a simple principle: **identify and remove everything that is *not* the real background**. What remains must be the leak.



A MORE ADVANCED ATTACK: SEEING THROUGH THE SOFTWARE'S OWN EYES

Instead of building a separate system to remove the virtual background, this "meet-in-the-middle" attack re-purposes the video conference software's *own matting algorithm* against itself. This results in a cleaner and more effective extraction of leaked pixels.



This method recovers **over 53% more leaked pixel information** than previous approaches, revealing clearer objects, text, and environmental details.

Source: Weissberg et al., "Seeing Through"

The very tools designed to create the virtual background can be turned into tools to dismantle it.

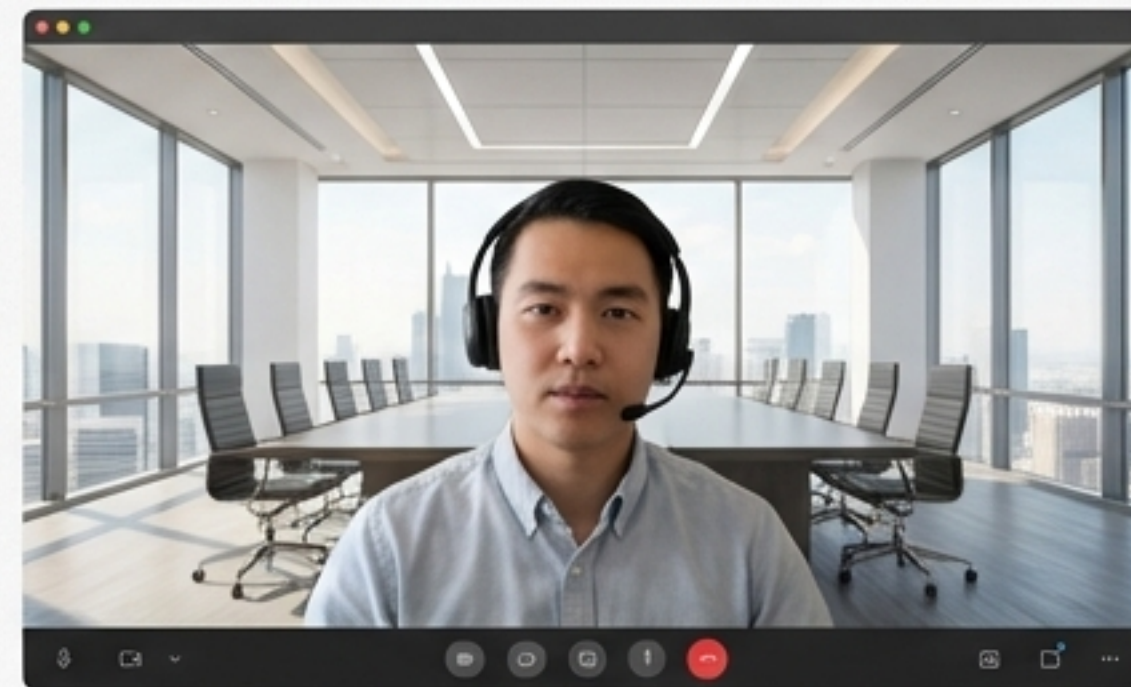
A NEW LINE OF INQUIRY: CAN WE DETECT A FORGERY?

The Scenario: The threat isn't just about hiding a messy room. A malicious actor could use a virtual background for active deception—for example, pretending to be in a secure office during a sensitive call when they are actually in a public space.

The Forensic Question
Can we distinguish a real, camera-captured background from a digitally inserted virtual one?



Claim: Real Background



Claim: Real Background

Research by Nowroozi et al. provides a method to forensically detect these forgeries.

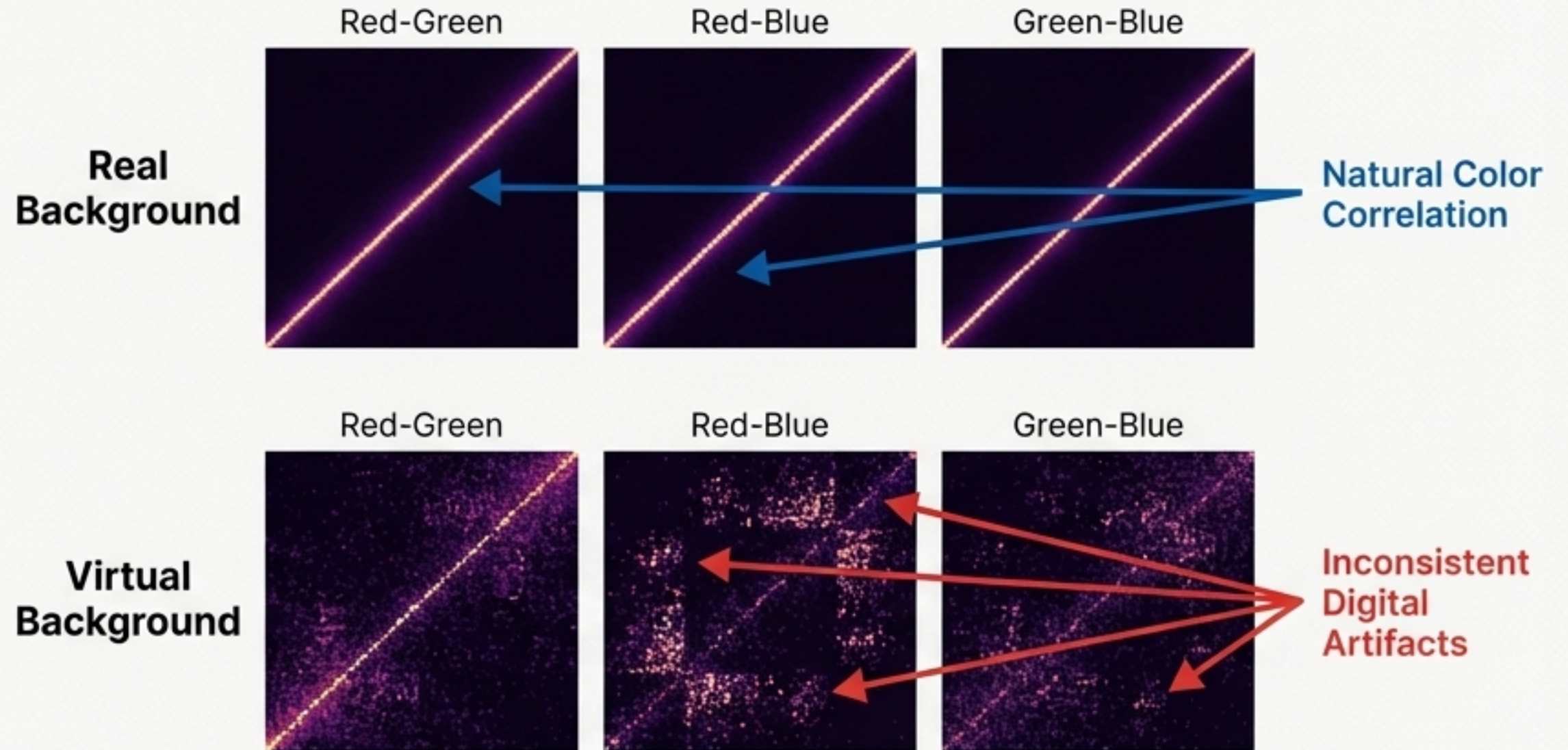
REAL AND VIRTUAL BACKGROUNDS LEAVE DISTINCT STATISTICAL FINGERPRINTS

The Method

Detection relies on analyzing co-occurrence matrices of color channels.

Real camera-captured scenes have natural correlations between Red, Green, and Blue channels.

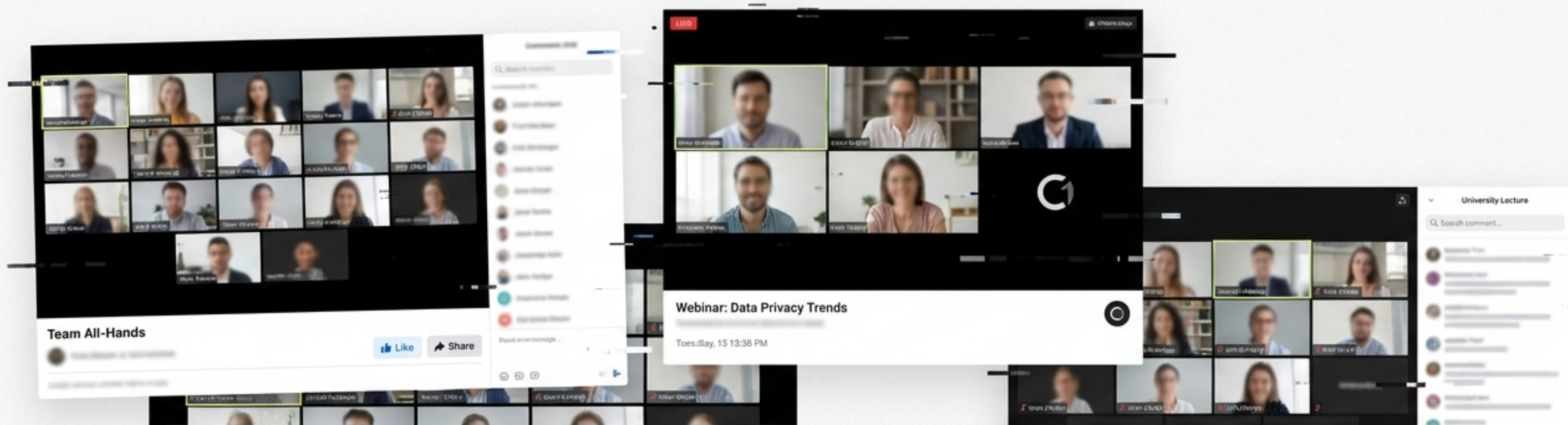
Virtual backgrounds, being digitally composed, lack these subtle statistical links.



By analyzing the statistical relationships *between* color channels (spectral co-occurrence), a detector can distinguish real from virtual backgrounds with **~99% accuracy**.

THE ATTACK SURFACE EXTENDS BEYOND YOUR OWN SCREEN

The privacy threat isn't limited to what an active participant can extract from a live video feed. It extends to the common practice of sharing screenshots of meetings on social media and corporate websites.

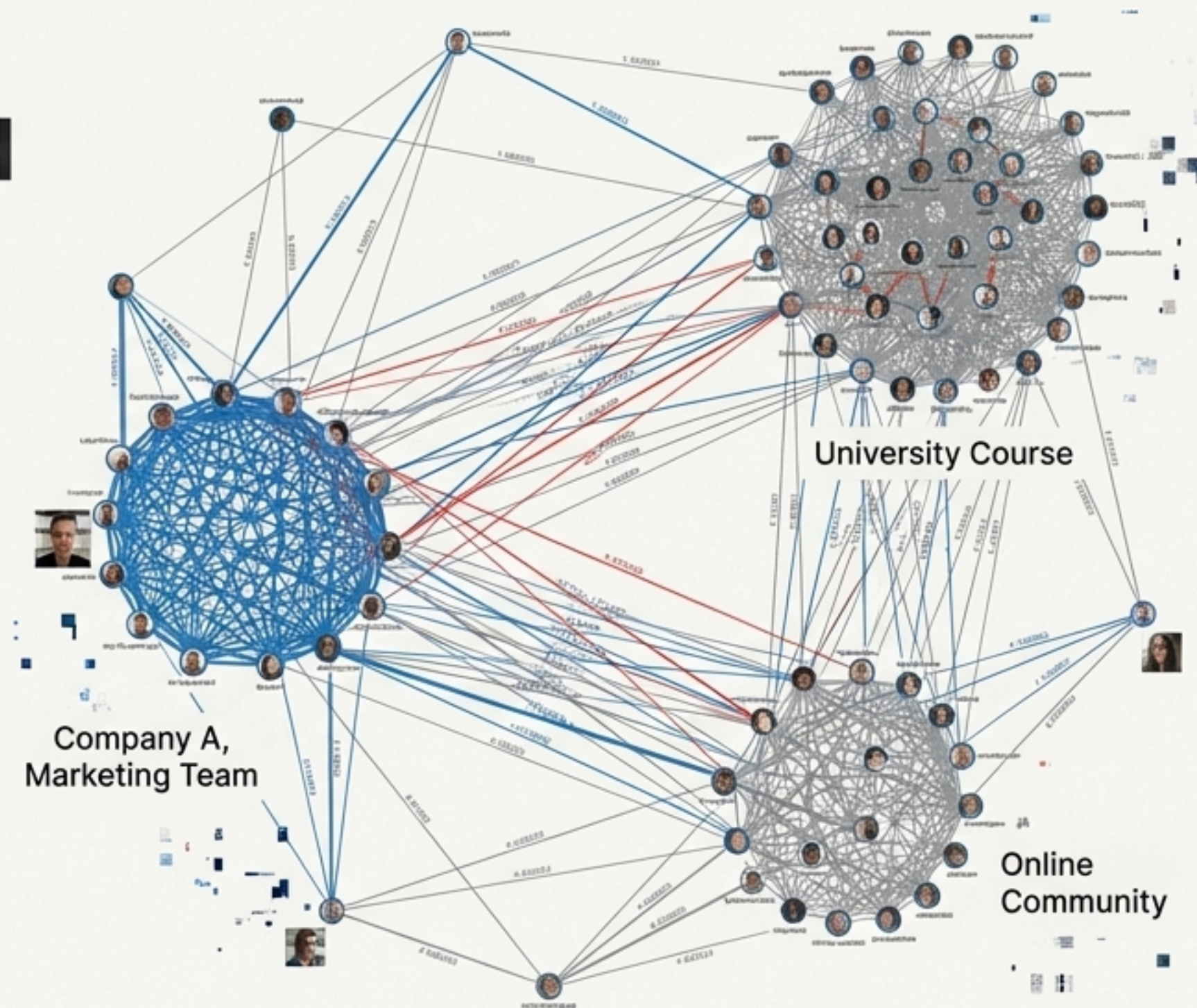


These seemingly innocuous images of team meetings, webinars, and online classes create a massive, publicly accessible, and highly structured dataset of personal information.

FROM PUBLIC COLLAGE TO PRIVATE SOCIAL GRAPH

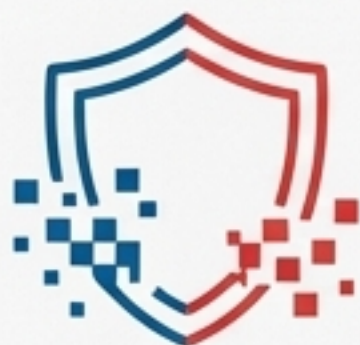
The Attack Flow

-  **Scrape:** Automatically collect thousands of meeting screenshots from social media sites like Twitter and Instagram.
-  **Extract:** Use machine learning to detect and extract faces, usernames, and estimate age and gender for every participant.
-  **Cross-Reference:** Link individuals across different meetings using facial recognition and username matching.



Publicly shared meeting screenshots can be used to map social and professional networks, creating privacy risks for **all participants**—even those who never shared an image themselves. An attacker can infer who works with whom, who attends which events, and build a detailed profile of an individual's connections.

THREAT SUMMARY: A MULTI-LAYERED RISK PROFILE



Technical Leakage

Imperfect, low-resolution segmentation inherently leaks real background pixels at the foreground/background boundary.



Active Exploitation

Leaked pixels can be aggregated and reconstructed, revealing objects, text, and sensitive details from a user's private environment.



Forensic Detection

Virtual backgrounds are forensically detectable with high accuracy, undermining their use for active deception (e.g., faking a location).



Social Data Leakage

Publicly shared meeting screenshots expose PII and can be used to map the social and professional graphs of all participants.

POTENTIAL TECHNICAL MITIGATIONS INVOLVE TRADE-OFFS



Dynamic Virtual Backgrounds

Dynamically adapt the brightness, saturation, and hue of the virtual background to mimic the properties of the real background, confusing pixel analysis. (Source: Sabra et al.)

Trade-off:

May reduce the visual consistency and quality of the virtual background.



Mask Erosion

A simple, efficient fix that shrinks the segmentation mask by a few pixels, cropping the user's silhouette slightly but removing most of the blended-pixel leak area. (Source: Weissberg et al.)

Trade-off:

Sacrifices visual quality by creating a harsh "cut-out" effect around the user.



Precise Segmentation

Segment the video at its original, high resolution instead of a down-scaled version, nearly eliminating leaks caused by scaling. (Source: Weissberg et al.)

Trade-off:

Computationally expensive; not feasible on most current hardware without dedicated ML accelerators.

Effective defenses currently require a compromise between privacy, visual quality, and system performance.

RECOMMENDATIONS FOR USERS AND ORGANIZATIONS

For Individual Users



Assume No True Privacy: Treat your virtual background as a cosmetic filter, not a security control. Always curate your physical background.



Be Mindful of Reflections: Sensitive information can be reflected in glasses, shiny surfaces, or windows.



Use Generic Pseudonyms: Avoid using your full, real name, which makes cross-referencing with social media profiles trivial.



Never Share Meeting Screenshots Publicly: Protect your own privacy and that of your colleagues.

For Organizations & Developers



Establish Clear Policies: Create and communicate guidelines for video call privacy and the public sharing of meeting images.



Educate Employees: Train staff on the risks outlined in this presentation.



Procure with Privacy in Mind: When choosing conferencing software, evaluate privacy-preserving features and vendor commitment to security.



Invest in Better Matting: For developers, move towards higher-resolution segmentation, leveraging hardware acceleration where possible, to reduce the root cause of leakage.



Virtual backgrounds are a useful feature, not a robust security control.

True privacy in video conferencing requires intentional action, both in the physical world and the digital one.

Source Research

Sabra et al., "Background Buster: Peeking through Virtual Backgrounds in Online Video Calls"; Weissberg et al., "Seeing Through: Analyzing and Attacking Virtual Backgrounds in Video Calls"; Nowroozi et al., "Real or virtual: a video conferencing background manipulation-detection system"; Intimbencing Inter Regular; Kagan et al., "Zooming Into Video Conferencing Privacy"